

Phase-2 Submission

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1. Problem Statement

Stock market forecasting is a critical task in financial analytics due to the high stakes involved in investment decisions. The stock market is inherently **non-linear, volatile, and affected by multiple unpredictable factors** such as global news, economic indicators, and investor sentiment. Traditional models like ARIMA, although statistically robust, are limited in capturing these **temporal dependencies and dynamic shifts** in market behavior. To address this, the project redefines the problem as a **regression-based time series prediction** task, where the goal is to predict future stock prices based on historical trends. The problem is solved using **deep learning techniques**, especially LSTM networks, which are well-suited for handling sequential data due to their memory cells that retain historical context.

Solving this problem is important as it offers **automated, accurate forecasting tools** that can assist investors, analysts, and financial institutions in making better-informed decisions, ultimately reducing risk and maximizing returns.

2. Project Objectives

The key objectives of this project have evolved after deeper exploration of the dataset and practical considerations.

They are:

- **Build a deep learning model** using LSTM to predict future stock prices.
- Compare performance with traditional statistical models like ARIMA or Prophet.
- **Capture hidden patterns and dependencies** in time-series data.
- Implement feature engineering to enhance model accuracy.
- Use **evaluation metrics** such as RMSE, MAE, and R^2 to quantify performance.
- Optionally, **deploy a Streamlit-based web application** for real-time forecasting.
- Ensure model **interpretability and transparency** to aid decision-making.

3. Flowchart of the Project Workflow

Data Collection



Data Cleaning & Preprocessing



Exploratory Data Analysis (EDA)



Feature Engineering



Model Building (ARIMA, LSTM)



Model Evaluation



Visualization & Interpretation



Deployment (Streamlit App - Optional)

4. Data Description

- **Source:** Yahoo Finance API ([yfinance](#)), optionally NSE/BSE for Indian stocks.
- **Type:** Structured, time-series data.
- **Features:** Daily stock data including Open, High, Low, Close (OHLC), and Volume.
- **Target Variable:** Closing Price (or Adjusted Close).
- **Number of Records:** Depends on stock and duration; typically several thousand rows.
- **Static vs. Dynamic:** Static dataset; only historical data is used (no live feeds).
- **Additional Features:** Derived technical indicators such as RSI, MACD, Bollinger Bands.

This dataset structure supports time-series forecasting and modeling with sequential inputs for deep learning

5. Data Preprocessing

A crucial phase in model building, the following preprocessing steps were performed:

- **Missing Values:** Handled through forward/backward fill or removal of incomplete rows.
- **Date Conversion:** Ensured datetime consistency and sorted chronologically.

- **Normalization:** *Min-Max Scaling applied to price and volume features to standardize input for LSTM.*
- **Lag Features:** *Created lagged versions of target and features to help model learn sequential dependencies.*
- **Rolling Statistics:** *Generated moving averages, rolling mean and standard deviation to capture trends.*
- **Categorical Encoding:** *Not required, as data is numerical.*
- **Train-Test Split:** *Data was split temporally to preserve sequence integrity—typically 80% for training, 20% for testing.*

6. Exploratory Data Analysis (EDA)

EDA was conducted using both statistical summaries and visualizations:

Univariate Analysis:

- *Plotted histograms and line plots for stock prices and volumes.*
- *Observed volatility, seasonality, and cyclical behavior in price movements.*

Bivariate/Multivariate Analysis:

- *Correlation heatmaps to evaluate relationships between technical indicators and target variable.*
- *Time-series plots comparing multiple companies (optional).*
- *Volume vs. Price scatterplots to understand buying/selling pressure.*

Insights:

- *Clear trends and reversals were visible in moving average plots.*
- *Technical indicators like MACD and RSI showed significant influence on price momentum.*
- *Price volatility varies with market phases (bull/bear), emphasizing the need for dynamic models.*

7. Feature Engineering

The following transformations and new features were introduced to improve model performance:

- ***Lag Features:*** *Previous day's closing price, volume, and indicators.*
- ***Rolling Features:*** *7-day, 14-day, and 30-day moving averages.*
- ***Technical Indicators:***
 - ***RSI (Relative Strength Index)*** *for momentum.*

- *MACD (Moving Average Convergence Divergence) for trend detection.*
- *Bollinger Bands for price volatility.*
- *Feature Scaling: Applied where needed for LSTM input compatibility.*
- *Date Parts (optional): Extracted week-day or month-seasonality if found significant.*

Each feature was selected based on financial theory or observed influence during EDA.

8. Model Building

We implemented and compared the following models:

Baseline Models:

- *ARIMA: Good for short-term trend prediction, but limited by assumptions of stationarity and linearity.*
- *Prophet: Captures trend and seasonality; easier to use but lacks deep representation.*

Deep Learning Models:

- *LSTM (Long Short-Term Memory): Implemented using TensorFlow/Keras, capable of capturing long-term dependencies in sequential data.*

- **Model Architecture:** 2 LSTM layers, followed by Dense layers.
- **Loss Function:** Mean Squared Error (MSE).
- **Optimizer:** Adam.
- **Training Strategy:** Early stopping and dropout layers used to prevent overfitting.

Evaluation Metrics Used:

- **MAE** (Mean Absolute Error)
- **RMSE** (Root Mean Squared Error)
- **R² Score** (Coefficient of Determination)

LSTM significantly outperformed ARIMA in both RMSE and generalizability.

9. Visualization of Results & Model Insights

The following visualizations were created for interpretation and evaluation:

- **Actual vs. Predicted Plot:** Visual comparison showing the model's accuracy.
- **Residual Plots:** Checked for error patterns—residuals were normally distributed.

- **Feature Importance (using SHAP or attention scores):** Showed which indicators influenced predictions most.
- **Rolling Predictions:** Visualized model predictions over different time windows.
- **Training Loss Plot:** Ensured convergence and stability during training.

These visual insights helped validate the model and interpret how it behaves under various market conditions.

10. Tools and Technologies Used

- **Programming Language:** Python 3.8
- **IDE:** Google Colab
- **Libraries:**
 - **Data Handling:** pandas, numpy
 - **Time Series Analysis:** yfinance, statsmodels, fbprophet
 - **Deep Learning:** tensorflow, keras
 - **Visualization:** matplotlib, seaborn, plotly
 - **Deployment:** Streamlit (for local web interface)
- **Version Control:** GitHub (to be updated)

11. Team Members and Contributions

<i>Name</i>	<i>Responsibilities</i>
<i>Mohamed Saif</i>	<i>- Model Development (LSTM & ARIMA)- Deployment Planning (Streamlit)- Model Evaluation</i>
<i>Diya Angelin S.P.</i>	<i>- Data Cleaning- Feature Engineering (Lag, Rolling, Technical Indicators)</i>
<i>Naveen M</i>	<i>- Exploratory Data Analysis (EDA)- Visualization (Charts, Trends, Correlations)</i>
<i>Sanjay M</i>	<i>- Documentation and Report Writing- Preparing Final Submission- Presentation Materials</i>